PRODUCT USER MANUAL

For Atlantic -Iberian Biscay Irish- Ocean Physics Reanalysis Product

IBI_MULTIYEAR_PHY_005_002
Issue: 5.0

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Approval Date: November 2025

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2014/05/05	all	Creation of the document.	M. G. Sotillo A. Amo S. Cailleau	E. Álvarez Fanjul L. Crosnier
1.1	2015/05/01	all	Change format to fit Copernicus Marine Service graphical rules		L. Crosnier
2.0	2015/12/15	all	**First release** of the IBI-MFC **PHY-V2** MY hindcast. Temporal coverage fom 2002 to 2014.	M. G. Sotillo A. Amo B. Levier	M. G. Sotillo
3.0	2017/12/15	all	**[EIS March 2018: IBI-V4 version]** **First release** of the IBI-MFC MY **PHY/BGC-V4** product. Temporal coverage from 01/01/1992 to 31/12/2016.	M. G. Sotillo A. Amo B. Levier	M. G. Sotillo
3.1	2019/01/15		**[EIS April 2019: IBI-V4 version]** **Temporal extension:** from 01/01/2014 to 24/12/2017.	M. G. Sotillo A. Amo B. Levier	M. G. Sotillo
3.2	2019/04/08		**[EIS July 2019: IBI-V4 version]** **Temporal Extension from:** 01/01/2014 to 24/12/2018. New dataset for static files.	M. G. Sotillo A. Amo B. Levier	M. G. Sotillo
3.3	2020/04/03		**[EIS July 2020: IBI-V4 version]** **Temporal Extension from:** from 01/01/2018 to 24/12/2019.	M. G. Sotillo A. Amo B. Levier R. Aznar	M. G. Sotillo
4.0	2020/09/10		**[EIS December 2019: IBI-V5 version]** **First release** of the IBI-MFC MY **PHY/BGC-V5** product Modification of product/datasets names following new convention.	M. G. Sotillo A. Amo B. Levier R. Aznar	M. G. Sotillo
4.1	2022/08/08	all	**[EIS November 2022: IBI-V5 version]** **Temporal Extension:** from 11/10/2019 to 2020/12/28. Adaptation of the document to the new template. New yearly and climatological datasets.	A. Amo B. Levier R. Aznar	M. G. Sotillo
4.2	2024/05/31		**[EIS November 2024: IBI-V5 version]** Automatic temporal extension New 1/12° Interim dataset. New Variables: Vertical velocity and air-sea fluxes.	L. Castrillo R. Aznar B. Levier	S. Cailleau M.G. Sotillo

4.3	2025/03/28		**[Climatology files]** Metadata modification and addition of the new variable climatology_bounds.	L. Castrillo R. Aznar B. Levier	S. Cailleau S. Ciliberti
5.0	2025/05/28		**[EIS November 2025: IBI-V5 version]** First release of the IBI-MFC MY PHY/BGC-V6 product. Increased spatial resolution up to 1/36° Doodson filtered variables available (currents and ssh) Modification of datasets configuration following new convention.	L. Castrillo R. Aznar B. Levier S. Ciliberti	S. Cailleau S. Ciliberti

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GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
CMEMS	Copernicus Marine Service
ECMWF	European Centre for Medium Range Weather forecast
IBI	Iberian-Biscay-Ireland
INS	In situ
NRT	Near Real Time
MDS	Marine Data Store
MFC	Monitoring and Forecasting Centre
MY	Multi Year
NetCDF	Network Common Data Format
PUM	Product User Manual
QUID	Quality Information Document
SLA	Sea Level Anomaly
SST	Sea Surface Temperature
TAC	Thematic Assembly Centre

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the Product User Manual (PUM) for the biogeochemistry Multi-Year (MY) product **IBI_MULTIYEAR_PHY_005_002** which provide continuous data from **01/01/1993 until M-4**. Datasets are composed of reanalysis data as soon as available and completed with interim system files meanwhile.

This product is defined as a standard grid at **1/36 degree**, (approx. 2-3 km), extending from 19°W to 5°E and 26°N to 56°N.

The product is organized in the following datasets:

1. **cmems_mod_ibi_phy-cur_my_0.027deg_(PT1H-m / P1D-m / P1M-m / P1Y-m):** mean fields of ocean currents.
2. **cmems_mod_ibi_phy-temp_my_0.027deg_(PT1H-m / P1D-m / P1M-m / P1Y-m):** mean fields of temperature.
3. **cmems_mod_ibi_phy-ssh_my_0.027deg_(PT1H-m / P1D-m / P1M-m / P1Y-m):** mean fields of significant wave height.
4. **cmems_mod_ibi_phy-mld_my_0.027deg_(PT1H-m / P1D-m / P1M-m / P1Y-m):** mean fields of mixed layer depth.
5. **cmems_mod_ibi_phy-sal_my_0.027deg_(P1D-m / P1M-m / P1Y-m):** mean fields of salinity.
6. **cmems_mod_ibi_phy-wcur_my_0.027deg_(P1D-m / P1M-m / P1Y-m):** mean fields of vertical velocities.
7. **cmems_mod_ibi_phy-ssh_my_detided-0.027deg_(P1D-m / P1M-m / P1Y-m):** mean fields doodson detided sea surface height.
8. **cmems_mod_ibi_phy-cur_my_detided-0.027deg_(P1D-m / P1M-m / P1Y-m):** mean fields of doodson detided currents.
9. **cmems_mod_ibi_phy-mflux_my_0.027deg_(P1D-m / P1M-m / P1Y-m):** 2D mean fields of momentum fluxes.
10. **cmems_mod_ibi_phy-wflux_my_0.027deg_(P1D-m / P1M-m / P1Y-m):** 2D mean fields of water fluxes.
11. **cmems_mod_ibi_phy-hflux_my_0.027deg_(P1D-m / P1M-m / P1Y-m):** 2D mean fields of heat fluxes.
12. **cmems_mod_ibi_phy_my_0.027deg-3D_static:** bathymetry, masks, mean dynamic topography and coordinates files used to generate this IBI product.
13. **cmems_mod_ibi_phy_my_0.027deg-climatology_P1M-m:** climatological parameters (monthly mean and standard deviation) of the variables delivered in the daily resolution (except air-sea fluxes dataset), for the period 1993-2016, and climatology_bounds.

The IBI-MFC Production Unit, run by NOW Systems with the support of CESGA (that provides needed supercomputing resources) is responsible for the generation and delivery of the product, while the model setup and the generation of the native products are managed by Mercator Ocean International.

The product is available on-line and disseminated through the Copernicus Marine Data Store after automatic and human quality controls. Files downloaded are in **NetCDF-4** format and follow **CF-1.8** convention.
The IBI-PHY MY modelling system and validation are extensively described in the Quality Information Document, while a short quality overview of qualified Essential Ocean Variables is provided in the Synthesis Quality Overview: both documents are accessible through the dedicated IBI-PHY MY product page.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

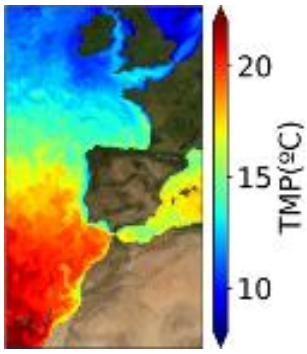
Product Lines	**IBI_MULTIYEAR_PHY_005_002**
Geographical coverage	IBI : 19°W-5°E; 26°N – 56°N
Variables	• Potential temperature • Eastward velocity • Northward velocity • Zonal barotropic velocity • Meridional barotropic velocity • Sea surface height • Ocean mixed layer thickness defined by density • Sea floor potential temperature • Salinity • Upward sea water velocity • E-R-P • Precipitation • Wind stress module • Downward heat flux • Net downward shortwave radiation flux • Net downward longwave radiation flux • Downward sensible heat flux • Downward latent heat flux • climatology_bounds. • Bathymetry • Mask • Coordinates (e1t, e2t, e3t) • Mean dynamic topography
Product Type	Reanalysis - Interim
Available time series	01/01/1993 – M-4
Temporal resolution	Daily – Monthly – Yearly means
Update Frequency	Monthly
Target delivery time	/

Horizontal resolution	1/36 degree
Number of vertical levels	50
Format	NetCDF4

b) Production System Description

The IBI-MFC provides an ocean physical MY product for the Iberia-Biscay-Ireland (IBI) area starting in **01/01/1993** and being regularly updated on a monthly basis. The model system is developed and run by Mercator-Ocean international for generating the core timeseries from 1993 and the periodic temporal extensions, being the Interim system and product post-processed to the user's format by NOW Systems with the support of CESGA in terms of supercomputing resources.

The IBI model numerical core used for both the multi-year and interim products is identical and is based on the NEMO v3.6 ocean general circulation model run at 1/36° horizontal resolution. Altimeter data, in situ temperature and salinity vertical profiles and satellite sea surface temperature are assimilated.

Domain **Resolution and grid** **Geographic coverage**	IBI (19°W-5°E; 26°N – 56°N) 1/36° (2-3 km); regular grid; 1081 x 865 This product is regional. It is defined on a regular grid at 1/36 degree (approx. 2-3 km) that goes from 26°N to 56°N in latitude and 19°W to 5°E in longitude. Latitude and longitude step is 0.02778°. The resulting horizontal grid extends to 1081x865 grid points. 
System Name	IBI36-PHY-MY_v6
Model version	NEMO v3.6 (**Madec et al., 2017**)
Horizontal resolution	1/36°

Vertical resolution	50 z*-levels
Interpolation Method	Only horizontal interpolation: bilinear interpolation
Coupling	Online coupling with IBI-BGC MY model application.
Atmospheric forcings	Hourly ECMWF Reanalysis ERA5 (10-m wind, surface pressure, 2-m temperature, relative humidity, precipitations, short-wave and long-wave radiative fluxes).
Initial conditions & boundary conditions	Global Ocean Physical Multi-Year product GLOBAL_MULTIYEAR_PHY_001_030 at 1/12° resolution
Rivers	33-point sources Merge of daily Copernicus Marine INS data and climatological data from GRDC and Banque Hydro dataset. 2D fluxes are additionally corrected using coastal runoff-rates (on a monthly basis) from **Dai and Trenberth (2002)** climatology.
Tides	Yes (11 tidal components, astro pot)
Bathymetry	GEBCO08 (**Becker et al., 2009**) merged with several local databases (F. Lyard, personal communication, 2010).
Free surface	Explicit, non-linear, time-splitting
Vertical mixing	k-epsilon
Tracer advection	QUICKEST scheme (**Leonard, 1979**) with the limiter proposed by **Zalesak (1979)** for the advection component.
Surge capability	Yes
Ocean colour effects	OCEANCOLOUR_GLO_BGC_L4_MY_009_104 and OCEANCOLOUR_GLO_BGC_L4_NRT_009_102 Copernicus Marine products - before 1998, monthly climatology 1998-2019 - from 1998, monthly values
Assimilation scheme	MROA2 (previously known as SAM2 (SEEK Filter)) + IAU + bias correction
Assimilated observations	- Along-track satellite Sea Level Anomaly from Copernicus Marine Service SLA-TAC (SEALEVEL_EUR_PHY_L3_MY_008_061, SEALEVEL_EUR_PHY_L3_NRT_008_059) - In situ temperature and salinity vertical profiles from Copernicus Marine INS-TAC (CORA 5.2 database,

	INSITU_GLO_TS_ASSIM_REP_OBSERVATIONS_013_051). - Satellite Sea Surface Temperature from Copernicus Marine SST-TAC (SST_GLO_PHY_L3S_MY_010_039).

c) Dataset details

Product Lines	IBI_MULTIYEAR_PHY_005_002
Geographical coverage	IBI

Datasets:

cmems_mod_ibi_phy-cur_my_0.027deg_PT1H-m
cmems_mod_ibi_phy-cur_my_0.027deg_P1D-m
cmems_mod_ibi_phy-cur_my_0.027deg_P1M-m
cmems_mod_ibi_phy-cur_my_0.027deg_P1Y-m

cmems_mod_ibi_phy-temp_my_0.027deg_PT1H-m
cmems_mod_ibi_phy-temp_my_0.027deg_P1D-m
cmems_mod_ibi_phy-temp_my_0.027deg_P1M-m
cmems_mod_ibi_phy-temp_my_0.027deg_P1Y-m

cmems_mod_ibi_phy-ssh_my_0.027deg_PT1H-m
cmems_mod_ibi_phy-ssh_my_0.027deg_P1D-m
cmems_mod_ibi_phy-ssh_my_0.027deg_P1M-m
cmems_mod_ibi_phy-ssh_my_0.027deg_P1Y-m

cmems_mod_ibi_phy-mld_my_0.027deg_PT1H-m
cmems_mod_ibi_phy-mld_my_0.027deg_P1D-m
cmems_mod_ibi_phy-mld_my_0.027deg_P1M-m
cmems_mod_ibi_phy-mld_my_0.027deg_P1Y-m

cmems_mod_ibi_phy-sal_my_0.027deg_P1D-m
cmems_mod_ibi_phy-sal_my_0.027deg_P1M-m
cmems_mod_ibi_phy-sal_my_0.027deg_P1Y-m

cmems_mod_ibi_phy-wcur_my_0.027deg_P1D-m
cmems_mod_ibi_phy-wcur_my_0.027deg_P1M-m
cmems_mod_ibi_phy-wcur_my_0.027deg_P1Y-m

cmems_mod_ibi_phy-ssh_my_detided-0.027deg_P1D-m
cmems_mod_ibi_phy-ssh_my_detided-0.027deg_P1M-m
cmems_mod_ibi_phy-ssh_my_detided-0.027deg_P1Y-m

cmems_mod_ibi_phy-cur_my_detided-0.027deg_P1D-m
cmems_mod_ibi_phy-cur_my_detided-0.027deg_P1M-m
cmems_mod_ibi_phy-cur_my_detided-0.027deg_P1Y-m

cmems_mod_ibi_phy-mflux_my_0.027deg_P1D-m
cmems_mod_ibi_phy-mflux_my_0.027deg_P1M-m
cmems_mod_ibi_phy-mflux_my_0.027deg_P1Y-m

cmems_mod_ibi_phy-wflux_my_0.027deg_P1D-m
cmems_mod_ibi_phy-wflux_my_0.027deg_P1M-m
cmems_mod_ibi_phy-wflux_my_0.027deg_P1Y-m

cmems_mod_ibi_phy-hflux_my_0.027deg_P1D-m cmems_mod_ibi_phy-hflux_my_0.027deg_P1M-m cmems_mod_ibi_phy-hflux_my_0.027deg_P1Y-m cmems_mod_ibi_phy_my_0.027deg-3D_static cmems_mod_ibi_phy_my_0.027deg-climatology_P1M-m		
Product Type	Reanalysis	Interim
Update frequency	Yearly	Monthly
Available time series	01/01/1993 – Interim	Reanalysis – M-4
Target delivery time	/	/
Temporal resolution	Hourly mean Daily mean Monthly mean Yearly mean	

cmems_mod_ibi_phy-cur_my_0.027deg_PT1H-m contains the 2D hourly mean fields
uo [m/s] Eastward velocity *eastward_sea_water_velocity*
vo [m/s] Northward velocity *northward_sea_water_velocity*
ubar [m/s] zonal barotropic velocity *barotropic_eastward_sea_water_velocity*
vbar [m/s] meridional barotropic velocity *barotropic_northward_sea_water_velocity*
cmems_mod_ibi_phy-cur_my_0.027deg_P1D-m **cmems_mod_ibi_phy-cur_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-cur_my_0.027deg_P1Y-m** contain 3D mean fields
uo [m/s] Eastward velocity *eastward_sea_water_velocity*
vo [m/s] Northward velocity *northward_sea_water_velocity*
cmems_mod_ibi_phy-temp_my_0.027deg_PT1H-m contains 2D hourly mean fields
thetao [°C]

Temperature *sea_water_potential_temperature*
cmems_mod_ibi_phy-temp_my_0.027deg_P1D-m **cmems_mod_ibi_phy-temp_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-temp_my_0.027deg_P1Y-m** contain 3D mean fields of potential tempature and 2D mean fields of bottom temperature
thetao [°C] Temperature sea_water_potential_temperature
bottomT [°C] Sea floor potential temperature *sea_water_potential_temperature_at_sea_floor*
cmems_mod_ibi_phy-ssh_my_0.027deg_PT1H-m **cmems_mod_ibi_phy-ssh_my_0.027deg_P1D-m** **cmems_mod_ibi_phy-ssh_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-ssh_my_0.027deg_P1Y-m** contain 2D mean fields
zos [m] Sea surface height *sea_surface_height_above_geoid*
cmems_mod_ibi_phy-mld_my_0.027deg_PT1H-m **cmems_mod_ibi_phy-mld_my_0.027deg_P1D-m** **cmems_mod_ibi_phy-mld_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-mld_my_0.027deg_P1Y-m** contain 3D mean fields
mlotst [m] Ocean mixed layer thickness defined by density *ocean_mixed_layer_thickness_defined_by_sigma_theta*
cmems_mod_ibi_phy-sal_my_0.027deg_P1D-m **cmems_mod_ibi_phy-sal_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-sal_my_0.027deg_P1Y-m** contain 3D mean fields
so [PSU] Salinity *sea_water_salinity*
cmems_mod_ibi_phy-wcur_my_0.027deg_P1D-m **cmems_mod_ibi_phy-wcur_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-wcur_my_0.027deg_P1Y-m** contain 3D mean fields

wo [m/s] Upward sea water velocity *upward_sea_water_velocity*
cmems_mod_ibi_phy-ssh_my_detided-0.027deg_P1D-m **cmems_mod_ibi_phy-ssh_my_detided-0.027deg_P1M-m** **cmems_mod_ibi_phy-ssh_my_detided-0.027deg_P1Y-m** contain 2D mean fields
zos_detided [m] Sea surface height assuming no tide *sea_surface_height_above_geoid_assuming_no_tide*
cmems_mod_ibi_phy-cur_my_detided-0.027deg_P1D-m **cmems_mod_ibi_phy-cur_my_detided-0.027deg_P1M-m** **cmems_mod_ibi_phy-cur_my_detided-0.027deg_P1Y-m** contain 2D mean fields
uo_detided [m/s] Eastward velocity assuming no tide *eastward_sea_water_velocity_assuming_no_tide*
vo_detided [m/s] Northward velocity assuming no tide *northward_sea_water_velocity_assuming_no_tide*
cmems_mod_ibi_phy-wflux_my_0.027deg_P1D-m **cmems_mod_ibi_phy-wflux_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-wflux_my_0.027deg_P1Y-m** contain 2D mean fields
sowaflup [kg m-2 s-1] Water flux out of sea ice and sea water *water_flux_out_of_sea_ice_and_sea_water*
sowaprec [kg m-2 s-1] Precipitation flux *precipitation_flux*
cmems_mod_ibi_phy-mflux_my_0.027deg_P1D-m **cmems_mod_ibi_phy-mflux_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-mflux_my_0.027deg_P1Y-m** contain 2D mean fields
sotausrf [N m-2] Magnitude of surface downward stress *magnitude_of_surface_downward_stress*
cmems_mod_ibi_phy-hflux_my_0.027deg_P1D-m **cmems_mod_ibi_phy-hflux_my_0.027deg_P1M-m** **cmems_mod_ibi_phy-hflux_my_0.027deg_P1Y-m** contain 2D mean fields
soheflds [W m-2] surface downward heat flux in sea water *surface_downward_heat_flux_in_sea_water*
soceshwf [W m-2] Net downward shortwave flux at sea water surface *net_downward_shortwave_flux_at_sea_water_surface*
socelowf [W m-2] Surface net downward longwave flux

surface_net_downward_longwave_flux
socesenf [W m-2] Surface downward sensible heat flux *surface_downward_sensible_heat_flux*
socelatf [W m-2] surface downward latent heat flux *surface_downward_latent_heat_flux*
cmems_mod_ibi_phy_my_0.027deg-climatology_P1M-m contain 3D and 2D mean fields Monthly mean and standard deviation of all the variables delivered in the daily resolution datasets **except static**
Climatology bounds [hours since time reference] climatology_bounds time interval endpoints
cmems_mod_ibi_phy_my_0.027deg-3D_static
IBI-MFC_005_002_mask_bathy.nc **deptho [m]** Bathymetry sea_floor_depth_below_geoid **mask [-]** Land-sea mask: 1 = sea; 0 = land sea_binary_mask
IBI-MFC_005_002_coordinates.nc **e1t [m]** Cell dimension along X axis **e2t [m]** Cell dimension along Y axis **e3t [m]** Cell dimension along Z axis
IBI-MFC_005_002_mdt.nc **mdt [m]** Mean dynamic topography *sea_surface_height_above_geoid*

d) Additional information on some parameters

mlotst [m]	ocean_mixed_layer_thickness_defined_by_sigma_theta. It is the depth where the density increase compared to density at 10 m depth corresponds to a temperature decrease of 0.01°C in local surface conditions (Θ10m, S10m, P0= 0 db, surface pressure).
zos[m]	sea_surface_height_above_geoid. The geoid is a surface of constant geopotential with which mean sea level would coincide if the ocean were at rest. The parameter "ssh" is the difference between the actual sea surface height at any given time and place, and that which it

	would have if the ocean were at rest.

e) Processing information

IBI-PHY MY product processing is performed according to the following schedule:

- IBI-PHY MY **Reanalysis** timeseries is produced, in native format, by Mercator Ocean International at the availability of the upstream data and then transferred to NOW Systems for the overall transformation according to standard formats. The production is performed once per year.
- IBI-PHY MY **Interim** datasets are operationally produced by NOW Systems on monthly basis, starting from the last available reanalysis date and covering a period up to M-4.

f) Catalogue updating

Reanalysis data are uploaded to the MDS as soon as the production of the reanalysis timeseries in native format and its post-processing are fully completed. **Reanalysis** files will overwrite the **Interim** ones for a corresponding date as soon as it is possible to reprocess the corresponding temporal period. The **Interim** system produces **Interim** datasets on a monthly basis ensuring MY time series availability until M-4.

g) Time averaging

Dataset frequency	Time averaging	Forecast lead time
montlhy	monthly means over the calendar month (first to last day of the month centered at mid of the month)	/
hourly	hourly means over an hour (centered at mid hour).	
daily	daily means over a day (midnight to midnight, centered at noon).	
yearly	yearly mean over the calendar year (first to last day of the year centered at the mid of the year)	

h) Product use and impact

The IBI-PHY MY product provides high resolution specialized past reconstruction of the physical conditions in the Atlantic Iberian-Biscay-Irish region, with added value in terms of modelling setup and spatial resolution with respect to the solution available the same area from the Global Ocean Physical MY one. Because of its specificities and complete catalogue, that provides information on Essential Ocean Variables from hourly to yearly frequency, the IBI-PHY MY is tailored to support specific sectors of Blue Economy such as:

- **Climate and Adaptation**: physical ocean variables such as ocean currents, sea level, temperature, salinity, are fundamental variables to understand the

contribution to climate change and support the national & local authorities, governmental & public environmental agencies in addressing recommendations for the IPCC in the Europe.

- **Extremes, hazards and safety**: physical variables are directly related to potential impacts when reaching extreme values such as Marine Heat Waves. Long records such us the IBI-PHY-MY product are crucial to characterize extremes for better monitoring as well as in support to planning of emergency response activities (for example, by analyzing the impact of plastic pollution, perform search and rescue response, etc.).
- **Ocean Health**: thanks to the understanding of trends of physical processes such as ocean currents or the mixed layer depth, it is possible to improve the monitoring capacity of high-risk coastal areas.
- **Trade & Marine Navigation**: the IBI-PHY MY datasets like ocean currents can valuably support applications in the Maritime sector such as route optimization, safety planning and environmental monitoring.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

CMEMS_{fileVersion}_{region}_PHY_MY_${productionCenter}_${freqFlag}_{datatype}_{validDate}_{validDate}_R{bulletinDate}_{productType}.nc

Where:

- **fileVersion**: vxry, where x, y are the version and release number, respectively.
- **region**: a three-letter code for the region, IBI in this case.
- **productionCenter**: a letter code for the production center (NL = Nologin Oceanic Weather Systems).
- **freqFlag:** the frequency of data values in the file (01hav = hourly averaged, 01dav = daily averaged, 01mav = monthly averaged, 01yav = yearly averaged).
- **datatype (optional):** it is a parameter which informs about the kind of included parameters.
- **validDate:** YYYYMMDD is the valid date of the fields contained in the file. For the monthly and yearly datasets, this field is the initial and final date of the month and year, respectively.

- **bulletinDate:** RYYYYMMDD is the bulletin date, i.e.: data production date (e.g.: R20201201)
- **productType:** is a two-letter+ two-number code for the product type (RE01 for reanalysis and IN01 for interim).

Examples:
From the hourly dataset request:
CMEMS_v6r1_IBI_PHY_MY_NL_01hav_19990101_19990101_R20201201_RE01.nc

Climatology dataset.
File nomenclature, when downloaded through the FTP service, is as follows:
CMEMS_${fileVersion}_${region}_PHY_MYCLIM_1993_2016_${month}.nc
Where:
- **region:** a three-letter code for the region, IBI in this case.
- **fileVersion:** vxry, where x, y are the version and release number, respectively
- **month:** 01 to 12 for January to December climatology.

Examples:
CMEMS_v6r1_IBI_PHY_MYCLIM_1993_2016_04.nc

5) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_mod_ibi_phy-cur_my_0.027deg_PT1H-m	CMEMS_v6r1_IBI_PHY_MY_NL_01hav_cur_19950101_19950101_R19950104_RE01.nc	0.2
cmems_mod_ibi_phy-cur_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_cur_19950101_19950101_R19950104_RE01.nc	0.15
cmems_mod_ibi_phy-temp_my_0.027deg_PT1H-m	CMEMS_v6r1_IBI_PHY_MY_NL_01hav_temp_19950101_19950101_R19950104_RE01.nc	0.05
cmems_mod_ibi_phy-temp_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_temp_19950101_19950101_R19950104_RE01.nc	0.05
cmems_mod_ibi_phy-ssh_my_0.027deg_PT1H-m	CMEMS_v6r1_IBI_PHY_MY_NL_01hav_ssh_19950101_19950101_R19950104_RE01.nc	0.05
cmems_mod_ibi_phy-ssh_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_ssh_19950101_19950101_R19950104_RE01.nc	0.01
cmems_mod_ibi_phy-mld_my_0.027deg_PT1H-m	CMEMS_v6r1_IBI_PHY_MY_NL_01hav_mld_19950101_19950101_R19950104_RE01.nc	0.05
cmems_mod_ibi_phy-mld_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_mld_19950101_19950101_R19950104_RE01.nc	0.01
cmems_mod_ibi_phy-sal_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_sal_19950101_19950101_R19950104_RE01.nc	0.05
cmems_mod_ibi_phy-wcur_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_wcur_19950101_19950101_R19950104_RE01.nc	0.1
cmems_mod_ibi_phy-ssh_my_detided-0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_sshDetided_19950101_19950101_R19950104_RE	0.01

	01.nc	
cmems_mod_ibi_phy-cur_my_detided-0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_curDetided_19950101_19950101_R19950104_RE01.nc	0.01
cmems_mod_ibi_phy-mflux_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_mflux_19950101_19950101_R19950104_RE01.nc	0.01
cmems_mod_ibi_phy-wflux_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_wflux_19950101_19950101_R19950104_RE01.nc	0.01
cmems_mod_ibi_phy-hflux_my_0.027deg_(P1D-m/P1M-m/P1Y-m)	CMEMS_v6r1_IBI_PHY_MY_NL_01dav_hflux_19950101_19950101_R19950104_RE01.nc	0.01
cmems_mod_ibi_phy_my_0.027deg-climatology_P1M-m	CMEMS_v6r1_IBI_PHY_MYCLIM_1993_2016_03.nc	0.5
cmems_mod_ibi_phy_my_0.027deg-3D_static	IBI-MFC_005_002_coordinates.nc IBI-MFC_005_002_mask_bathy.nc IBI-MFC_005_002_mdt.nc	0.3

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in the **Annex**, for each dataset.

7) REFERENCES

Becker, J. J., Sandwell, D. T., Smith, W. H. F., Braud, J., B. Binder, B., Depner, J., Fabre, D., Factor, J., Ingalls, S., Kim, S.-H., Ladner, R., Marks, K., Nelson, S., Pharaoh, A., Trimmer, R., Von Rosenberg, J., Wallace, G., and Weatherall, P. (2009). Global bathymetry and elevation data at 30 arc seconds resolution: SRTM30 PLUS. Mar Geod., 32, 355-371. https://doi.org/10.1080/01490410903297766

Dai, A., and Trenberth K.E. (2002). Estimates of Freshwater Discharge from Continents: Latitudinal and Seasonal Variations. J. of Hydrometeorology, 3, 660-687. https://doi.org/10.1175/1525-7541(2002)003<0660:EOFDFC>2.0.CO;2

Leonard, B. P. (1979). A stable and accurate convective modelling procedure based on quadratic upstream interpolation, Comp. Method. Appl. M. 19, 59–98.

Madec, G. and the NEMO Team (2017). NEMO ocean engine. Note du Pôle de modélisation, Institut Pierre-Simon Laplace (IPSL), France, No 27, ISSN No 1288-1619. Available at https://zenodo.org/records/3248739 (last access: 28/05/2025).

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Zalesak, S. T. 1979: Fully multidimensional flux-corrected transport algorithms for fluids, J. Computat. Phys., 31, 335–362.

8) ANNEX

cmems_mod_ibi_phy-cur_my_0.027deg_PT1H-m

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01hav_cur_19950101_19950101_R19950104_RE01 {
dimensions:
	time = 24 ;
	longitude = 871 ;
	latitude = 1078 ;
variables:
	float time(time) ;
		time:standard_name = "time" ;
		time:long_name = "time" ;
		time:axis = "T" ;
		time:units = "hours since 1950-01-01" ;
		time:calendar = "gregorian" ;
	float longitude(longitude) ;
		longitude:standard_name = "longitude" ;
		longitude:long_name = "Longitude" ;
		longitude:units = "degrees_east" ;
		longitude:unit_long = "Degrees East" ;
		longitude:step = "0.08333f" ;
		longitude:axis = "X" ;
	float latitude(latitude) ;
		latitude:standard_name = "latitude" ;
		latitude:long_name = "Latitude" ;
		latitude:units = "degrees_north" ;
		latitude:unit_long = "Degrees North" ;
		latitude:step = "0.08333f" ;
		latitude:axis = "Y" ;
	float ubar(time, latitude, longitude) ;
		ubar:_FillValue = 9.96921e+36f ;
		ubar:standard_name = "barotropic_eastward_sea_water_velocity" ;
		ubar:long_name = "zonal barotropic velocity" ;
		ubar:units = "m s-1" ;
		ubar:unit_long = "Meters per second" ;
		ubar:easting = "longitude" ;
		ubar:northing = "latitude" ;
		ubar:longitude_min = "-19.f" ;
		ubar:longitude_max = "5.f" ;
		ubar:latitude_min = "26.f" ;
		ubar:latitude_max = "56.f" ;
	float vbar(time, latitude, longitude) ;
		vbar:_FillValue = 9.96921e+36f ;
		vbar:standard_name = "barotropic_northward_sea_water_velocity" ;
		vbar:long_name = "meridional barotropic velocity" ;
		vbar:units = "m s-1" ;
		vbar:unit_long = "Meters per second" ;
		vbar:easting = "longitude" ;
		vbar:northing = "latitude" ;
		vbar:longitude_min = "-19.f" ;
		vbar:longitude_max = "5.f" ;
		vbar:latitude_min = "26.f" ;
		vbar:latitude_max = "56.f" ;
	float uo(time, latitude, longitude) ;
		uo:_FillValue = 9.96921e+36f ;
		uo:standard_name = "eastward_sea_water_velocity" ;
		uo:long_name = "Eastward velocity" ;
		uo:units = "m s-1" ;
		uo:unit_long = "Meters per second" ;
		uo:easting = "longitude" ;
		uo:northing = "latitude" ;
		uo:longitude_min = "-19.f" ;
		uo:longitude_max = "5.f" ;
		uo:latitude_min = "26.f" ;
		uo:latitude_max = "56.f" ;
	float vo(time, latitude, longitude) ;
```

```
            vo:_FillValue = 9.96921e+36f ;
            vo:standard_name = "northward_sea_water_velocity" ;
            vo:long_name = "Northward velocity" ;
            vo:units = "m s-1" ;
            vo:unit_long = "Meters per second" ;
            vo:easting = "longitude" ;
            vo:northing = "latitude" ;
            vo:longitude_min = "-19.f" ;
            vo:longitude_max = "5.f" ;
            vo:latitude_min = "26.f" ;
            vo:latitude_max = "56.f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean surface hourly mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-cur_my_0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_cur_19950101_19950101_R19950104_RE01 {
dimensions:
      time = 1 ;
      longitude = 871 ;
      latitude = 1078 ;
      depth = 50 ;
variables:
      float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.08333f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.08333f" ;
            latitude:axis = "Y" ;
      float depth(depth) ;
            depth:standard_name = "depth" ;
            depth:long_name = "Depth" ;
            depth:units = "m" ;
            depth:unit_long = "Meters" ;
            depth:positive = "down" ;
            depth:axis = "Z" ;
      float uo(time, depth, latitude, longitude) ;
            uo:_FillValue = 9.96921e+36f ;
            uo:standard_name = "eastward_sea_water_velocity" ;
            uo:long_name = "Eastward velocity" ;
            uo:units = "m s-1" ;
```

```
            uo:unit_long = "Meters per second" ;
            uo:easting = "longitude" ;
            uo:northing = "latitude" ;
            uo:longitude_min = "-19.f" ;
            uo:longitude_max = "5.f" ;
            uo:latitude_min = "26.f" ;
            uo:latitude_max = "56.f" ;
            uo:z_min = "0.50576f" ;
            uo:z_max = "5698.061f" ;
      float vo(time, depth, latitude, longitude) ;
            vo:_FillValue = 9.96921e+36f ;
            vo:standard_name = "northward_sea_water_velocity" ;
            vo:long_name = "Northward velocity" ;
            vo:units = "m s-1" ;
            vo:unit_long = "Meters per second" ;
            vo:easting = "longitude" ;
            vo:northing = "latitude" ;
            vo:longitude_min = "-19.f" ;
            vo:longitude_max = "5.f" ;
            vo:latitude_min = "26.f" ;
            vo:latitude_max = "56.f" ;
            vo:z_min = "0.50576f" ;
            vo:z_max = "5698.061f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-temp_my_0.027deg_PT1H-m

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01hav_temp_19950101_19950101_R19950104_RE01 {
dimensions:
      time = 24 ;
      longitude = 871 ;
      latitude = 1078 ;
variables:
      float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.08333f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.08333f" ;
            latitude:axis = "Y" ;
      float thetao(time, latitude, longitude) ;
```

```
            thetao:_FillValue = 9.96921e+36f ;
            thetao:standard_name = "sea_water_potential_temperature" ;
            thetao:long_name = "Temperature" ;
            thetao:units = "degrees_C" ;
            thetao:unit_long = "degrees_C" ;
            thetao:easting = "longitude" ;
            thetao:northing = "latitude" ;
            thetao:longitude_min = "-19.f" ;
            thetao:longitude_max = "5.f" ;
            thetao:latitude_min = "26.f" ;
            thetao:latitude_max = "56.f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean surface hourly mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-temp_my_0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_temp_19950101_19950101_R19950104_RE01 {
dimensions:
      time = 1 ;
      longitude = 871 ;
      latitude = 1078 ;
      depth = 50 ;
variables:
      float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.08333f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.08333f" ;
            latitude:axis = "Y" ;
      float depth(depth) ;
            depth:standard_name = "depth" ;
            depth:long_name = "Depth" ;
            depth:units = "m" ;
            depth:unit_long = "Meters" ;
            depth:positive = "down" ;
            depth:axis = "Z" ;
      float thetao(time, depth, latitude, longitude) ;
            thetao:_FillValue = 9.96921e+36f ;
            thetao:standard_name = "sea_water_potential_temperature" ;
            thetao:long_name = "Temperature" ;
            thetao:units = "degrees_C" ;
```

```
            thetao:unit_long = "degrees_C" ;
            thetao:easting = "longitude" ;
            thetao:northing = "latitude" ;
            thetao:longitude_min = "-19.f" ;
            thetao:longitude_max = "5.f" ;
            thetao:latitude_min = "26.f" ;
            thetao:latitude_max = "56.f" ;
            thetao:z_min = "0.50576f" ;
            thetao:z_max = "5698.061f" ;
      float bottomT(time, latitude, longitude) ;
            bottomT:_FillValue = 9.96921e+36f ;
            bottomT:standard_name = "sea_water_potential_temperature_at_sea_floor" ;
            bottomT:long_name = "Sea floor potential temperature" ;
            bottomT:units = "degrees_C" ;
            bottomT:unit_long = "degrees_C" ;
            bottomT:easting = "longitude" ;
            bottomT:northing = "latitude" ;
            bottomT:longitude_min = "-19.f" ;
            bottomT:longitude_max = "5.f" ;
            bottomT:latitude_min = "26.f" ;
            bottomT:latitude_max = "56.f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-ssh_my_0.027deg_(PT1H-m / P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_ssh_19950101_19950101_R19950104_RE01 {
dimensions:
      time = 1 ;
      longitude = 871 ;
      latitude = 1078 ;
variables:
      float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.08333f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.08333f" ;
            latitude:axis = "Y" ;
      float zos(time, latitude, longitude) ;
            zos:_FillValue = 9.96921e+36f ;
            zos:standard_name = "sea_surface_height_above_geoid" ;
```

```
            zos:long_name = "Sea surface height" ;
            zos:units = "m" ;
            zos:unit_long = "Meters" ;
            zos:easting = "longitude" ;
            zos:northing = "latitude" ;
            zos:longitude_min = "-19.f" ;
            zos:longitude_max = "5.f" ;
            zos:latitude_min = "26.f" ;
            zos:latitude_max = "56.f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-mld_my_0.027deg_(PT1H-m / P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_mld_19950101_19950101_R19950104_RE01 {
dimensions:
      time = 1 ;
      longitude = 871 ;
      latitude = 1078 ;
variables:
      float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.08333f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.08333f" ;
            latitude:axis = "Y" ;
      float mlotst(time, latitude, longitude) ;
            mlotst:_FillValue = 9.96921e+36f ;
            mlotst:standard_name = "ocean_mixed_layer_thickness_defined_by_sigma_theta" ;
            mlotst:long_name = "Ocean mixed layer thickness defined by density" ;
            mlotst:units = "m" ;
            mlotst:unit_long = "Meters" ;
            mlotst:easting = "longitude" ;
            mlotst:northing = "latitude" ;
            mlotst:longitude_min = "-19.f" ;
            mlotst:longitude_max = "5.f" ;
            mlotst:latitude_min = "26.f" ;
            mlotst:latitude_max = "56.f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
```

```
            :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-sal_my_0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_sal_19950101_19950101_R19950104_RE01 {
dimensions:
      time = 1 ;
      longitude = 871 ;
      latitude = 1078 ;
      depth = 50 ;
variables:
      float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.08333f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.08333f" ;
            latitude:axis = "Y" ;
      float depth(depth) ;
            depth:standard_name = "depth" ;
            depth:long_name = "Depth" ;
            depth:units = "m" ;
            depth:unit_long = "Meters" ;
            depth:positive = "down" ;
            depth:axis = "Z" ;
      float so(time, depth, latitude, longitude) ;
            so:_FillValue = 9.96921e+36f ;
            so:standard_name = "sea_water_salinity" ;
            so:long_name = "Salinity" ;
            so:units = "1e-3" ;
            so:unit_long = "Practical Salinity Unit" ;
            so:easting = "longitude" ;
            so:northing = "latitude" ;
            so:longitude_min = "-19.f" ;
            so:longitude_max = "5.f" ;
            so:latitude_min = "26.f" ;
            so:latitude_max = "56.f" ;
            so:z_min = "0.50576f" ;
            so:z_max = "5698.061f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
```

```
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-wcur_my_0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_wcur_19950101_19950101_R19950104_RE01 {
dimensions:
      time = 1 ;
      longitude = 871 ;
      latitude = 1078 ;
      depth = 50 ;
variables:
      float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.08333f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.08333f" ;
            latitude:axis = "Y" ;
      float depth(depth) ;
            depth:standard_name = "depth" ;
            depth:long_name = "Depth" ;
            depth:units = "m" ;
            depth:unit_long = "Meters" ;
            depth:positive = "down" ;
            depth:axis = "Z" ;
      float wo(time, depth, latitude, longitude) ;
            wo:_FillValue = 9.96921e+36f ;
            wo:standard_name = "upward_sea_water_velocity" ;
            wo:long_name = "quasi geostrophic vertical velocity" ;
            wo:units = "m s-1" ;
            wo:unit_long = "Meters per second" ;
            wo:easting = "longitude" ;
            wo:northing = "latitude" ;
            wo:longitude_min = "-19.f" ;
            wo:longitude_max = "5.f" ;
            wo:latitude_min = "26.f" ;
            wo:latitude_max = "56.f" ;
            wo:z_min = "0f" ;
            wo:z_max = "5800f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
```

```
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-ssh_my_detided-0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_sshDetided_19950101_19950101_R19950104_RE01 {
dimensions:
      time = 1 ;
      longitude = 871 ;
      latitude = 1078 ;
variables:
      float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
            time:axis = "T" ;
            time:units = "hours since 1950-01-01" ;
            time:calendar = "gregorian" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.08333f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.08333f" ;
            latitude:axis = "Y" ;
      float zos_detided(time, latitude, longitude) ;
            zos_detided:_FillValue = 9.96921e+36f ;
            zos_detided:standard_name = "sea_surface_height_above_geoid_assuming_no_tide" ;
            zos_detided:long_name = "sea surface height above geoid assuming no tide" ;
            zos_detided:units = "m" ;
            zos_detided:unit_long = "Meters" ;
            zos_detided:easting = "longitude" ;
            zos_detided:northing = "latitude" ;
            zos_detided:longitude_min = "-19.f" ;
            zos_detided:longitude_max = "5.f" ;
            zos_detided:latitude_min = "26.f" ;
            zos_detided:latitude_max = "56.f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```

cmems_mod_ibi_phy-cur_my_detided-0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_curDetided_19950101_19950101_R19950104_RE01 {
dimensions:
	time = 1 ;
	longitude = 871 ;
	latitude = 1078 ;
variables:
	float time(time) ;
		time:standard_name = "time" ;
		time:long_name = "time" ;
		time:axis = "T" ;
		time:units = "hours since 1950-01-01" ;
		time:calendar = "gregorian" ;
	float longitude(longitude) ;
		longitude:standard_name = "longitude" ;
		longitude:long_name = "Longitude" ;
		longitude:units = "degrees_east" ;
		longitude:unit_long = "Degrees East" ;
		longitude:step = "0.08333f" ;
		longitude:axis = "X" ;
	float latitude(latitude) ;
		latitude:standard_name = "latitude" ;
		latitude:long_name = "Latitude" ;
		latitude:units = "degrees_north" ;
		latitude:unit_long = "Degrees North" ;
		latitude:step = "0.08333f" ;
		latitude:axis = "Y" ;
	float uo_detided(time, latitude, longitude) ;
		uo_detided:_FillValue = 9.96921e+36f ;
		uo_detided:standard_name = "northward_sea_water_velocity_assuming_no_tide" ;
		uo_detided:long_name = "northward sea water velocity assuming no tide" ;
		uo_detided:units = "m s-1" ;
		uo_detided:unit_long = "Meters per second" ;
		uo_detided:easting = "longitude" ;
		uo_detided:northing = "latitude" ;
		uo_detided:longitude_min = "-19.f" ;
		uo_detided:longitude_max = "5.f" ;
		uo_detided:latitude_min = "26.f" ;
		uo_detided:latitude_max = "56.f" ;
	float vo_detided(time, latitude, longitude) ;
		vo_detided:_FillValue = 9.96921e+36f ;
		vo_detided:standard_name = "eastward_sea_water_velocity_assuming_no_tide" ;
		vo_detided:long_name = "eastward sea water velocity assuming no tide" ;
		vo_detided:units = "m s-1" ;
		vo_detided:unit_long = "Meters per second" ;
		vo_detided:easting = "longitude" ;
		vo_detided:northing = "latitude" ;
		vo_detided:longitude_min = "-19.f" ;
		vo_detided:longitude_max = "5.f" ;
		vo_detided:latitude_min = "26.f" ;
		vo_detided:latitude_max = "56.f" ;

// global attributes:
		:Conventions = "CF-1.8" ;
		:title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
		:source = "NEMO3.6" ;
		:domain_name = "IBI36" ;
		:field_type = "mean" ;
		:field_date = "19950101" ;
		:forecast_range = "01-day_reanalysis" ;
		:forecast_type = "reanalysis" ;
		:institution = "NOW Systems (Spain)" ;
		:references = "http://marine.copernicus.eu/" ;
		:contact = "https://marine.copernicus.eu/contact" ;
		:licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
		:comment = "" ;
}
```

cmems_mod_ibi_phy-mflux_my_0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_mflux_19950101_19950101_R19950104_RE01 {
dimensions:
	time = 1 ;
	longitude = 871 ;
	latitude = 1078 ;
variables:
	float time(time) ;
		time:standard_name = "time" ;
		time:long_name = "time" ;
		time:axis = "T" ;
		time:units = "hours since 1950-01-01" ;
		time:calendar = "gregorian" ;
	float longitude(longitude) ;
		longitude:standard_name = "longitude" ;
		longitude:long_name = "Longitude" ;
		longitude:units = "degrees_east" ;
		longitude:unit_long = "Degrees East" ;
		longitude:step = "0.08333f" ;
		longitude:axis = "X" ;
	float latitude(latitude) ;
		latitude:standard_name = "latitude" ;
		latitude:long_name = "Latitude" ;
		latitude:units = "degrees_north" ;
		latitude:unit_long = "Degrees North" ;
		latitude:step = "0.08333f" ;
		latitude:axis = "Y" ;
	float taum(time, latitude, longitude) ;
		taum:_FillValue = 9.96921e+36f ;
		taum:standard_name = "magnitude_of_surface_downward_stress" ;
		taum:long_name = "Magnitude of surface downward stress" ;
		taum:units = "N m-2" ;
		taum:unit_long = "Newton per square meter" ;
		taum:easting = "longitude" ;
		taum:northing = "latitude" ;
		taum:longitude_min = "-19.f" ;
		taum:longitude_max = "5.f" ;
		taum:latitude_min = "26.f" ;
		taum:latitude_max = "56.f" ;

// global attributes:
		:Conventions = "CF-1.8" ;
		:title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
		:source = "NEMO3.6" ;
		:domain_name = "IBI36" ;
		:field_type = "mean" ;
		:field_date = "19950101" ;
		:forecast_range = "01-day_reanalysis" ;
		:forecast_type = "reanalysis" ;
		:institution = "NOW Systems (Spain)" ;
		:references = "http://marine.copernicus.eu/" ;
		:contact = "https://marine.copernicus.eu/contact" ;
		:licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
		:comment = "" ;
}
```

cmems_mod_ibi_phy-wflux_my_0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_wflux_19950101_19950101_R19950104_RE01 {
dimensions:
	time = 1 ;
	longitude = 871 ;
	latitude = 1078 ;
variables:
	float time(time) ;
		time:standard_name = "time" ;
		time:long_name = "time" ;
		time:axis = "T" ;
		time:units = "hours since 1950-01-01" ;
		time:calendar = "gregorian" ;
```

```
        float longitude(longitude) ;
                longitude:standard_name = "longitude" ;
                longitude:long_name = "Longitude" ;
                longitude:units = "degrees_east" ;
                longitude:unit_long = "Degrees East" ;
                longitude:step = "0.08333f" ;
                longitude:axis = "X" ;
        float latitude(latitude) ;
                latitude:standard_name = "latitude" ;
                latitude:long_name = "Latitude" ;
                latitude:units = "degrees_north" ;
                latitude:unit_long = "Degrees North" ;
                latitude:step = "0.08333f" ;
                latitude:axis = "Y" ;
        float wfo(time, latitude, longitude) ;
                wfo:_FillValue = 9.96921e+36f ;
                wfo:standard_name = "water_flux_into_sea_water" ;
                wfo:long_name = "water flux into sea water" ;
                wfo:units = "kg m-2 s-1" ;
                wfo:unit_long = "Kilogram per square meter and second" ;
                wfo:easting = "longitude" ;
                wfo:northing = "latitude" ;
                wfo:longitude_min = "-19.f" ;
                wfo:longitude_max = "5.f" ;
                wfo:latitude_min = "26.f" ;
                wfo:latitude_max = "56.f" ;
        float pr(time, latitude, longitude) ;
                pr:_FillValue = 9.96921e+36f ;
                pr:standard_name = "precipitation_flux" ;
                pr:long_name = "Precipitation flux" ;
                pr:units = "kg m-2 s-1" ;
                pr:unit_long = "Kilogram per square meter and second" ;
                pr:easting = "longitude" ;
                pr:northing = "latitude" ;
                pr:longitude_min = "-19.f" ;
                pr:longitude_max = "5.f" ;
                pr:latitude_min = "26.f" ;
                pr:latitude_max = "56.f" ;

// global attributes:
                :Conventions = "CF-1.8" ;
                :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
                :source = "NEMO3.6" ;
                :domain_name = "IBI36" ;
                :field_type = "mean" ;
                :field_date = "19950101" ;
                :forecast_range = "01-day_reanalysis" ;
                :forecast_type = "reanalysis" ;
                :institution = "NOW Systems (Spain)" ;
                :references = "http://marine.copernicus.eu/" ;
                :contact = "https://marine.copernicus.eu/contact" ;
                :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
                :comment = "" ;
}
```

cmems_mod_ibi_phy-hflux_my_0.027deg_(P1D-m / P1M-m / P1Y-m)

```
netcdf CMEMS_v6r1_IBI_PHY_MY_NL_01dav_hflux_19950101_19950101_R19950104_RE01 {
dimensions:
        time = 1 ;
        longitude = 871 ;
        latitude = 1078 ;
variables:
        float time(time) ;
                time:standard_name = "time" ;
                time:long_name = "time" ;
                time:axis = "T" ;
                time:units = "hours since 1950-01-01" ;
                time:calendar = "gregorian" ;
```

```
float longitude(longitude) ;
        longitude:standard_name = "longitude" ;
        longitude:long_name = "Longitude" ;
        longitude:units = "degrees_east" ;
        longitude:unit_long = "Degrees East" ;
        longitude:step = "0.08333f" ;
        longitude:axis = "X" ;
float latitude(latitude) ;
        latitude:standard_name = "latitude" ;
        latitude:long_name = "Latitude" ;
        latitude:units = "degrees_north" ;
        latitude:unit_long = "Degrees North" ;
        latitude:step = "0.08333f" ;
        latitude:axis = "Y" ;
float hfds(time, latitude, longitude) ;
        hfds:_FillValue = 9.96921e+36f ;
        hfds:standard_name = "surface_downward_heat_flux_in_sea_water" ;
        hfds:long_name = "surface downward heat flux in sea water" ;
        hfds:units = "W m-2" ;
        hfds:unit_long = "Watt per square meter" ;
        hfds:easting = "longitude" ;
        hfds:northing = "latitude" ;
        hfds:longitude_min = "-19.f" ;
        hfds:longitude_max = "5.f" ;
        hfds:latitude_min = "26.f" ;
        hfds:latitude_max = "56.f" ;
float rsntds(time, latitude, longitude) ;
        rsntds:_FillValue = 9.96921e+36f ;
        rsntds:standard_name = "net_downward_shortwave_flux_at_sea_water_surface" ;
        rsntds:long_name = "Net downward shortwave flux at sea water surface" ;
        rsntds:units = "W m-2" ;
        rsntds:unit_long = "Watt per square meter" ;
        rsntds:easting = "longitude" ;
        rsntds:northing = "latitude" ;
        rsntds:longitude_min = "-19.f" ;
        rsntds:longitude_max = "5.f" ;
        rsntds:latitude_min = "26.f" ;
        rsntds:latitude_max = "56.f" ;
float rlds(time, latitude, longitude) ;
        rlds:_FillValue = 9.96921e+36f ;
        rlds:standard_name = "surface_net_downward_longwave_flux" ;
        rlds:long_name = "surface net downward longwave flux" ;
        rlds:units = "W m-2" ;
        rlds:unit_long = "Watt per square meter" ;
        rlds:easting = "longitude" ;
        rlds:northing = "latitude" ;
        rlds:longitude_min = "-19.f" ;
        rlds:longitude_max = "5.f" ;
        rlds:latitude_min = "26.f" ;
        rlds:latitude_max = "56.f" ;
float hfss(time, latitude, longitude) ;
        hfss:_FillValue = 9.96921e+36f ;
        hfss:standard_name = "surface_downward_sensible_heat_flux" ;
        hfss:long_name = "Surface downward sensible heat flux" ;
        hfss:units = "W m-2" ;
        hfss:unit_long = "Watt per square meter" ;
        hfss:easting = "longitude" ;
        hfss:northing = "latitude" ;
        hfss:longitude_min = "-19.f" ;
        hfss:longitude_max = "5.f" ;
        hfss:latitude_min = "26.f" ;
        hfss:latitude_max = "56.f" ;
float hfls(time, latitude, longitude) ;
        hfls:_FillValue = 9.96921e+36f ;
        hfls:standard_name = "surface_downward_latent_heat_flux " ;
        hfls:long_name = "Surface downward latent heat flux" ;
        hfls:units = "W m-2" ;
        hfls:unit_long = "Watt per square meter" ;
        hfls:easting = "longitude" ;
```

```
            hfls:northing = "latitude" ;
            hfls:longitude_min = "-19.f" ;
            hfls:longitude_max = "5.f" ;
            hfls:latitude_min = "26.f" ;
            hfls:latitude_max = "56.f" ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Ocean 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "19950101" ;
            :forecast_range = "01-day_reanalysis" ;
            :forecast_type = "reanalysis" ;
            :institution = "NOW Systems (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}
```